LU HONGZHI

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RESEARCH INTERESTS

AI & Sustainability: Lifecycle Assessment (LCA) of Generative AI, LLM Carbon Modeling, Green Computing Policy.
Atmospheric Science: Pollution Dispersion Modeling, Renewable Energy Grid Impact Forecasting.
Sustainable Development: Environmental Policy Analysis, Hydrogeology & Watershed Management.

EDUCATION BACKGROUND

Master of Science in Environmental Science
Universiti Sains Malaysia (USM), Penang, Malaysia | Oct 2025 – Present (Expected July 2026)
Advisor: Assoc. Prof. Dr. Mohamad Anuar Bin Kamaruddin
Status: Thesis in progress | Language of Instruction: English
Bachelor of Management (Major: Land Resource Management)
Shenyang Jianzhu University, Liaoning, China | Sep 2021 – Jul 2025
GPA: 3.27 / 5.0 **Core Courses:** Atmospheric Science, Environmental Law, Sampling Technology, Hydrogeology, Scientific Writing.

PUBLICATIONS & PATENTS

- **Lu, H.**, & Lu, H. (2026). Transitioning to Adaptive Ecosystems: The Role of Artificial Intelligence in Sustainable Environmental Management and Pollution Control. *Journal of Technology and Humanities*, 7(1). DOI: **https://doi.org/10.53797/jthkkss.v7i1.1.2026**
- **Lu, H.**, & Lu, H. (2026). Empirical Lifecycle Assessment of Generative AI Inference Carbon Emissions: Global Trends and the Malaysian 'Carbon Arbitrage' Risk (2023–2030). *Uniglobal Journal of Social Sciences and Humanities*, 5(17). DOI: **https://doi.org/10.53797/ujssh.v5i17.1.2026**
- **Lu, H.** (2023). Patent: A new type of land resource detector. *State Intellectual Property Office of P.R. China*. Patent No: ZL202130777758.7.

RESEARCH EXPERIENCE

Lifecycle Assessment of Generative AI Inference Carbon Emissions: Global Trends (2023-2025) and Malaysian Grid Impact Forecasting (2030) (Master's Thesis)
Universiti Sains Malaysia | Oct 2025 – Present
Conducted a bottom-up Attributional Life Cycle Assessment (A-LCA) of GenAI inference carbon footprints to quantify Malaysia's "Digital Carbon Arbitrage" risks and propose Carbon Usage Effectiveness (CUE) standards.
Research on the Comprehensive Carrying Capacity of Land in Dalian City (Bachelor's Thesis)
Shenyang Jianzhu University | Sep 2024 – Jun 2025
Assessed urban land carrying capacity using a 17-indicator Entropy Weight Method to formulate sustainable green infrastructure and layout optimization policies for Dalian.

PROJECT & SOFTWARE DEVELOPMENT EXPERIENCE

Eco Scan Monster (Sustainability Mobile Application)
Independent Developer | Penang, Malaysia | Oct 2025 – Present

Concept & AI-Assisted Development: Engineered a mobile application aimed at enhancing public environmental awareness. Leveraged AI coding assistants (e.g., Google Gemini, OpenAI Codex) to accelerate application logic development, UI/UX interface design, and cross-platform adaptation.

AI Integration: Integrated a local AI vision model to analyze user-uploaded photos of waste or eco-items, automatically assigning an environmental impact score (1-10) to guide proper recycling and disposal.

Gamification Design: Designed an interactive behavioral reinforcement system featuring a virtual pet that gains XP and evolves through user's responsible recycling actions. Implemented an in-app "Monster Mall" and milestone-based "Eco Certificates" to strongly incentivize sustainable habits.

CONFERENCES & ACADEMIC ACTIVITIES

- **Participant**, SEA Youth Sustainability Summit 2024 (Co-organized by WWF-Malaysia & SEAmply Sustainable) | *Kuala Lumpur, Malaysia* | Aug 2024

TECHNICAL SKILLS

Software & AI-Assisted Engineering: Proficient in using AI coding models (Google Gemini, OpenAI Codex) for rapid full-stack development of mobile apps, desktop software, and websites; AI-driven UI/UX design; Local AI vision model integration; Gamification mechanism design.
Modeling & Data Analysis: Python (Highly proficient in leveraging AI tools for automated data cleaning, analytical scripting, and advanced data processing), SPSS, Entropy Weight Method, Lifecycle Assessment (LCA), Excel (Advanced).
Environmental Tools: ArcGIS (Spatial Visualization), Air Pollution Dispersion Modeling, Environmental Sampling.
General Skills: Patent Writing, Technical Report Compilation, Feasibility Analysis.

HONORS & AWARDS

Third Prize, 16th "Challenge Cup" Liaoning Province College Students Extracurricular Academic and Scientific Works Competition (Dec 2023)
Excellence Award, "White Horse Cup" Debate Competition, Shenyang Jianzhu University (Dec 2021)
Protest Youth Volunteer Certificate & Typical Representative of Anti-epidemic, Shenyang Jianzhu University (Sep 2021)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Served as Vice President of the Host Club and Captain of the Debate Team, demonstrating strong leadership, teamwork, and public speaking skills.

Active participant in community service, including extensive volunteer work during pandemic relief efforts.